

Proof

THE ASSERTION has been made that mathematics is uniquely characterized by something known as "proof." The first proof in the history of mathematics is said to have been given by Thales of Miletus (600 B.C.). He proved that the diameter divides a circle into two equal parts. Now this is a statement which is so simple that it appears self evident. The genius of the act was to understand that a proof is possible and necessary. What makes mathematical proof more than mere pedantry is its application in situations where the statements made are far less transparent. In the opinion of some, the name of the mathematics game is proof; no proof, no mathematics. In the opinion of others, this is nonsense; there are many games in mathematics.

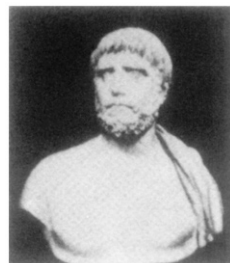
To discuss what proof is, how it operates, and what it is for, we need to have a concrete example of some complexity before us; and there is nothing better than to take a look at what undoubtedly is the most famous theorem in the history of mathematics as it appears in the most famous book in the history of mathematics. We allude to the Pythagorean Theorem, as it occurs in Proposition 47, Book I of Euclid's Elements (300 B.C.). We quote it in the English version given by Sir Thomas Heath. The in-text numbers at the right are references to previously established results or to "common notions."

Proposition 47 In right-angled triangles the square on the side subtending the right angle is equal to the squares on the sides containing the right angle.

Let ABC be a right-angled triangle having the angle BAC right;

I say that the square on BC is equal to the squares on BA, AC.

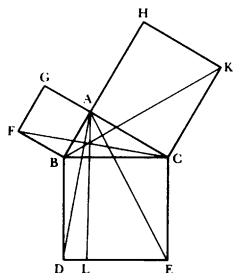
For let there be described on BC the square BDEC, and on BA, AC the squares GB, HC; [1.46] through A let AL



Thales
c. 624–548 B.C.



Pythagoras
c. 580–500 B.C.



be drawn parallel to either BD or CE, and let AD, FC be joined.

Then, since each of the angles BAC, BAG is right, it follows that with a straight line BA, and at the point A on it, the two straight lines AC, AG not lying on the same side make the adjacent angles equal to two right angles; therefore CA is in a straight line with AG. [1.14]

For the same reason BA is also in a straight line with AH.

And, since the angle DBC is equal to the angle FBA; for each is right: let the angle ABC be added to each; therefore the whole angle DBA is equal to the whole angle FBC. [C.N. 2]

And, since DB is equal to BC, and FB to BA, the two sides AB, BD are equal to the two sides FB, BC respectively, and the angle ABD is equal to the angle FBC: therefore the base AD is equal to the base FC, and the triangle ABD is equal to the triangle FBC. [1.4]

Now the parallelogram BL is double of the triangle ABD, for they have the same base BD and are in the same parallels BD, AL. [1.41]

And the square GB is double of the triangle FBC, for they again have the same base FB and are in the same parallels FB, GC. [1.41]

[But the doubles of equals are equal to one another.]

Therefore the parallelogram BL is also equal to the square GB.

Similarly, if AE, BK be joined, the parallelogram CL can also be proved equal to the square HC; therefore the whole square BDEC is equal to the two squares GB, HC. [C.N. 2]

And the square BDEC is described on BC, and the squares GB, HC on BA, AC.

Therefore the square on the side BC is equal to the squares on the sides BA, AC.

Therefore etc. Q.E.D.

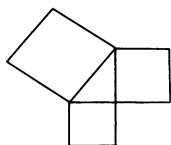
Now, assuming that we have read Euclid up to Proposition 47, and assuming we are able intellectually to get through this material, what are we to make of it all? Perhaps the most beautiful recorded reaction is that ascribed to Thomas Hobbes (1588–1679) by John Aubrey in his “Brief Lives”:

He was 40 yeares old before he looked on Geometry; which happened accidentally. Being in a Gentleman's Library, Euclid's Elements lay open, and 'twas the 47 *El. libri* I. He read the Proposition. By G—, sayd he (he would now and then swear an emphaticall Oath by way of emphasis) *this is impossible!* So he reads the Demonstration of it, which referred him back to such a Proposition; which proposition he read. That referred him back to another, which he also read. *Et sic deinceps* [and so on] that at last he was demonstratively convinced of that trueth. This made him in love with Geometry.

What appears initially as unintuitive, dubious, and somewhat mysterious ends up, after a certain kind of mental process, as gloriously true. Euclid, one likes to think, would have been proud of Hobbes and would use him as Exhibit A for the vindication of his long labors in compiling the Elements. Here is the proof process, discovered and promulgated by Greek mathematics, in the service of validation and certification. Now that the statement has been proved, we are to understand that the statement is true beyond the shadow of a doubt.

The backward referral to previous propositions, alluded to by Hobbes, is characteristic of the method of proof, and as we know, this can't go on forever. It stops with the so-called axioms and definitions. Whereas the latter are mere linguistic conventions, the former represent rock bottom self-evident facts upon which the whole structure is to rest, held together by the bolts of logic.

Also characteristic of the method is the considerable degree of abstraction that has occurred in the refinement of such concepts as triangle, right angle, square, etc. The figure itself appears here as a very necessary adjunct to the verbalization. In Euclid's presentation we cannot wholly follow the argumentation without the figure, and unless we are strong enough to imagine the figure in our mind's eye, we would be reduced to supplying our own figure if the author had not done it for us. Notice also that the language of the proof has a formal and severely restricted quality about it. This is not the language of history, nor of drama, nor of day to day life; this is language that has been sharpened



and refined so as to serve the precise needs of a precise but limited intellectual goal.

One response to this material was recorded by the poet Edna Millay, in her line, "Euclid alone has looked on Beauty bare." A shudder might even run down our spines if we believe that with a few magic lines of proof we have compelled all the right triangles in the universe to behave in a regular Pythagorean fashion.

Abstraction, formalization, axiomatization, deduction — here are the ingredients of proof. And the proofs in modern mathematics, though they may deal with different raw material or lie at a deeper level, have essentially the same feel to the student or the researcher as the one just quoted.

Further reading of Euclid's masterpiece brings up additional issues. Notice that in the figure certain lines, e.g. BK, AL seem to be extraneous to a minimal figure drawn as an expression of the theorem itself. Such a figure is illustrated here: a right angled triangle with squares drawn upon each of its three sides. The extraneous lines which in high school are often called "construction lines," complicate the figure, but form an essential part of the deductive process. They reorganize the figure into subfigures and the reasoning takes place precisely at this sublevel.

Now, how does one know where to draw these lines so as to reason with them? It would seem that these lines are accidental or fortuitous. In a sense this is true and constitutes the genius or the trick of the thing. Finding the lines is part of finding a proof, and this may be no easy matter. With experience come insight and skill at finding proper construction lines. One person may be more skillful at it than another. There is no guaranteed way to arrive at a proof. This sad truth is equally rankling to schoolchildren and to skillful professionals. Mathematics as a whole may be regarded as a systematization of just those questions which have been pursued successfully.

Mathematics, then, is the subject in which there are proofs. Traditionally, proof was first met in Euclid; and millions of hours have been spent in class after class, in country after country, in generation after generation, proving and reproving the theorems in Euclid. After the

introduction of the “new math” in the mid-nineteen fifties, proof spread to other high school mathematics such as algebra, and subjects such as set theory were deliberately introduced so as to be a vehicle for the axiomatic method and proof. In college, a typical lecture in advanced mathematics, especially a lecture given by an instructor with “pure” interests, consists entirely of definition, theorem, proof, definition, theorem, proof, in solemn and unrelieved concatenation. Why is this? If, as claimed, proof is validation and certification, then one might think that once a proof has been accepted by a competent group of scholars, the rest of the scholarly world would be glad to take their word for it and to go on. Why do mathematicians and their students find it worthwhile to prove again and yet again the Pythagorean theorem or the theorems of Lebesgue or Wiener or Kolmogoroff?

Proof serves many purposes simultaneously. In being exposed to the scrutiny and judgment of a new audience, the proof is subject to a constant process of criticism and revalidation. Errors, ambiguities, and misunderstandings are cleared up by constant exposure. Proof is respectability. Proof is the seal of authority.

Proof, in its best instances, increases understanding by revealing the heart of the matter. Proof suggests new mathematics. The novice who studies proofs gets closer to the creation of new mathematics. Proof is mathematical power, the electric voltage of the subject which vitalizes the static assertions of the theorems.

Finally, proof is ritual, and a celebration of the power of pure reason. Such an exercise in reassurance may be very necessary in view of all the messes that clear thinking clearly gets us into.

Further Readings. See Bibliography

R. Blanché; J. Dieudonné [1971]; G. H. Hardy; T. Heath; R. Wilder, [1944].